

Smart Rings for Early Detection of Peri-menopausal Depression

A Research Protocol

Janessa Griffith[†]
Information Science,
Cornell Tech, Cornell University,
NY, New York, USA
Janessa.griffith@cornell.edu

Tanzeem Choudhury
Information Science,
Cornell Tech, Cornell University,
NY, New York, USA
tanzeem.choudhury@cornell.edu

ABSTRACT

This protocol paper outlines a two-phased mixed methods study aimed at exploring the potential of smart rings for early detection of peri-menopausal depression. Peri-menopausal depression impacts approximately 40% of people during this life phase. Yet, few people are aware of this statistic, including clinicians. This lack of awareness has cascading impacts on seeking treatment. The aim of this research is to assess the utility of smart rings in the early detection of peri-menopausal depression.

In Phase 1, participants who are in the average age range of perimenopause (i.e., age 40 to 50) will wear as smart ring for three months. The ring will continuously monitor participants' sleep patterns and body temperature, providing sensor data. Concurrently, participants will maintain weekly journals.

Phase 2 involves interviews with participants to illuminate their experiences with the technology. These interviews will explore participant perceptions and any perceived benefits or limitations of using the smart ring.

By integrating sensor data with qualitative participant experiences, this study aims to provide a comprehensive understanding of the feasibility and utility of smart rings as a tool for early detection of peri-menopausal depression. Findings from this research could potentially inform future interventions and healthcare practices aimed at improving mental health outcomes during peri-menopause.

CCS CONCEPTS

• Ubiquitous Computing • Gender

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.
UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
© 2024 Copyright is held by the owner/author(s). Publication rights licensed to ACM.
ACM 979-8-4007-1058-2/24/10.
<https://doi.org/10.1145/3675094.3678426>

KEYWORDS

Smart ring, Wearables, gender, health

ACM Reference format:

Janessa Griffith and Tanzeem Choudhury. 2024. Smart Rings for Early Detection of Peri-menopausal Depression: A Research Protocol. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3678426>

1 Introduction

Peri-menopause is the phase in life where a person transitions to menopause (i.e., the cessation of menstrual cycles). Typical symptoms of peri-menopause include irregular menstrual cycles, hot flashes, night sweats, sleep disturbances, and mood changes, including depression [1,2]. Approximately 40% of women in peri-menopause will experience depression [3]. This risk is increased for women who have experienced depression at another time in their lives. Despite the prevalence, very few women will receive a diagnosis or seek treatment [4].

Depression in peri-menopause can be treated not only with therapy and antidepressants, but also with Hormone Replacement Therapy (HRT), or a combination of these treatments [5]. Thus, early detection of peri-menopausal depression has the potential to improve a person's quality of life. [6] note the seriousness of depression in peri-menopause, as one can be in this transitional phase for up to ten years, impacting one's quality of life. In fact, an Australian study identified that the highest rate of suicide for women was in the 45-49 age group [6], further compounding the critical need to address peri-menopausal depression.

The Oura Ring is a smart ring that tracks sleep stages (i.e., deep, REM, light) and offers sleep analyses, including sleep quality scores and insights. The ring also monitors heart rate, heart rate variability (HRV), activity levels, and body temperature, providing data on overall readiness and recovery [7]. Body temperature can serve as a proxy for estrogen levels; fluctuations in estrogen during the menstrual cycle or perimenopause can cause changes in body temperature [8].

The aim of the research is to assess the utility of smart rings (i.e., the Oura Ring) in detecting peri-menopause, thereby providing insights into why a person might be experiencing

depressive symptoms and what treatment options are available. The aim of this current paper is to highlight the research methods employed in this study, and garner feedback from peers in this field, improving the quality and robustness of this work. As very little research has been published at the intersection of digital health, peri-menopause, and mental health [9], this proposed research has the potential to offer new insights.

2 Methods

To address the research aim, we will take a mixed-methods approach in what Creswell and Plano-Clark [10] refer to as the Explanatory Sequential Design. In this design, quantitative data is collected and analyzed first, followed by qualitative data collection to explain or elaborate on the quantitative results. This approach is valuable for researchers who seek to understand their research findings by leveraging the strengths of both quantitative and qualitative approaches.

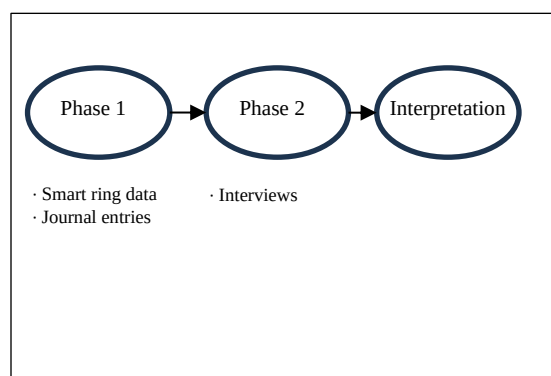


Figure 1: Explanatory Sequential Design starting with Phase 1 which will inform Phase 2. The methods are mixed in the interpretation phase.

2.1 Phase 1

The aim of the quantitative arm of this study is to assess the potential for the Oura Ring to produce a signal for early detection of peri-menopause. We expect this phase to take up to five months.

2.1.1 Study design.

This study arm will employ a prospective case series design. Participants will be asked to wear an Oura Ring for three months, continuously, and to journal their experiences. A case series design entails systematically collecting and presenting detailed data from a small group of peri-menopausal individuals using the Oura Ring and journal entries. This approach aims to describe how the smart rings monitor depressive symptoms, exploring potential trends and outcomes within this specific population. A case series is descriptive, focusing on the detailed observation and documentation of multiple cases with a similar condition. Therefore, a hypothesis is not used in this study. This study design

is well-suited as a preliminary investigation to assess the feasibility and effectiveness of using smart rings for mental health monitoring in this demographic, informing future research directions.

2.1.2 Participant recruitment and retention.

Participants will be recruited via social media. This will include Facebook, LinkedIn, Instagram, and X (formerly, Twitter). We will also hang posters on community boards in New York City. We will aim to recruit approximately 20 participants, as a pilot project. Participants will be given an honorarium for their participation in this study. The honorarium will be scaled according to ring wearing adherence to encourage participant retention.

2.1.3 Eligibility.

As the participants will need to be sized and given a ring, an eligibility criterion is that they will need to be in the New York City area (i.e., where the study is being conducted). Participants will be eligible if they self-identify as going through peri-menopause and experiencing depression or depressive symptoms and able to menstruate (even if their cycle is irregular). Those who have already reached menopause will be excluded from the study. Additionally, those on hormone replacement therapy, with diagnosed psychiatric disorders (other than depression), or with conditions that could affect sleep patterns (e.g., sleep apnea) will also be excluded from participating.

2.1.4 Data collection.

Participants will be asked to share their sleep, and menstrual cycle tracking data with the research team. Data will be collected using the participant's API token. Data will be anonymized using a participant identification number and stripped of any names or identifying information. Data will be stored on an institutionally managed Amazon Web Services (AWS) account. Participants will also be asked to keep a diary of reflections on their sleep, mood, and cycle.

2.1.5 Data analysis.

We will analyze the Oura Ring data on sleep and temperature (presented as daily data) and note any fluctuations. These daily measures will be compared against the weekly diaries kept by participants. For example, if a person notes feeling depressive symptoms, we will make observational notes about any sleep disturbances (e.g., to track vasomotor symptoms such as night sweats or hot flashes) or temperature fluctuations during that week.

2.2 Phase 2

To explain and expand upon findings from the quantitative arm of the study, we will conduct semi-structured interviews with participants from the first phase.

2.2.1 Study design.

A qualitative study design will be employed which involves systematically analyzing interview data to identify patterns, themes, and meanings within the content.

2.2.2 Participant recruitment.

Participants from the first phase will also be asked to participate in this second phase.

2.2.3 Data collection.

After a short demographic questionnaire, participants will be asked questions related to their use of the Oura Ring and whether it offered any insight into their peri-menopausal journey. These will include questions such as whether or not the participant has used digital tools in the past to detect or manage their depression during peri-menopause, strengths and weaknesses to using the Oura Ring, and suggestions for future digital tool development. The interview should last 30 to 60 minutes and will be audio-recorded (for later transcription and analysis).

2.2.4 Data analysis.

Audio-recordings will be transcribed and uploaded into qualitative analysis software. A thematic analysis will be performed, as described by Braun and Clarke [11]: the research team will start by thoroughly reading through the transcripts to become familiar with the data. Next, we will identify and code significant phrases and sentences, grouping these codes into themes that capture the essence of the data. Finally, we will review and refine these themes to ensure they accurately represent the underlying patterns and insights from the interviews.

2.3 Interpretation

In this explanatory sequential research design, we focused on perimenopausal individuals using smart rings and journal entries, the integration of qualitative participant interviews adds depth and context to the quantitative data gathered from the technology and personal diaries. The data from smart rings, capturing physiological and behavioral indicators of depression, combined with journal entries detailing subjective experiences and emotions during perimenopause, provide a comprehensive view of individuals' mental health journeys. Through qualitative interviews, researchers can explore nuances, such as personal coping strategies, social support networks, and the impact of specific life events on mental well-being, which may not be fully captured by quantitative measures alone. This mixed methods approach facilitates a richer interpretation by triangulating different sources of data, enhancing the understanding of how smart ring data and personal narratives intersect or diverge. It enables researchers to uncover complex patterns and individual variations in depression experiences during perimenopause, offering insights that neither method could achieve independently.

3 Discussion

A large proportion of people in perimenopause experience depression, making it critical to find ways to identify the start of this phase of life and seek appropriate care. This study protocol outlines the methods our research team plans to take to explore whether and how a smart ring could be used to detect early signs of peri-menopause. To address this research aim, we outlined a mixed-methods study protocol that leverages the strengths of both quantitative and qualitative methodologies, offering a more fulsome understanding of the intersection of perimenopausal depression and wearable technology.

Few studies focus on technological solutions for peri-menopausal depression [9]. Once executed, this study will add to the limited research in this area. It also serves as an example of how sex and

gender considerations can be incorporated into ubiquitous computing.

Using smart rings for detecting perimenopausal depression could significantly advance the literature on passive sensing by demonstrating how wearable technology can monitor physiological changes associated with mood disorders. By providing continuous, real-time data, smart rings could offer novel insights into the subtle physiological markers of mental health [12]. This research might also highlight the potential for integrating wearable technology into routine mental health monitoring, enhancing early detection and personalized treatment strategies [13]. Ultimately, the findings could enable broader applications of passive sensing technologies in understanding and managing various mental health conditions. Our participant sample will be from the greater New York City area; thus, our study population may not represent all people who experience perimenopausal depression. We are asking participants to self-identify as being in peri-menopause and experiencing depression or depressive symptoms. We recognize this as a limitation as not all people experiencing peri-menopause will know they are in this phase of life. Similarly, not all people will recognize they are experiencing depression or depressive symptoms. Additionally, we are asking participants to wear the smart ring for three months. This timeframe is short due to study limitations; however, observing fluctuating cycles during this timeframe might not be possible. We have a small sample size ($n=20$) for wearing the rings. This is due to budgetary restraints, as the smart rings start at \$299.00 USD. Another limitation, updates to the firmware could impact how the signals are processed and measured. If this occurs, we expect the updates to impact all study participants and will note changes, accordingly.

4 Conclusion

This protocol outlines a structured approach to explore the viability of smart rings for early detection of peri-menopausal depression. By integrating quantitative smart ring data with qualitative insights from participant journals and interviews, the study aims to provide a holistic understanding of smart ring technology's potential in detecting mental health during perimenopause. The findings could inform future interventions to improve women's mental health outcomes during this transitional phase. This research underscores the importance of innovative technological solutions in healthcare and highlights the value of mixed methods research in addressing complex health issues such as perimenopausal depression.

REFERENCES

- [1] Nancy Woods, Nina Coslov, Marcie Richardson, 2022. Perimenopause meets life: observations from the Women Living Better Survey. *Menopause*. 1;29(12), 1388-1398. <https://doi.org/10.1097/GME.0000000000002072>.
- [2] Claire Hardy and Myra Hunter, 2021. Premenstrual symptoms and work: Exploring female staff experiences and recommendations for workplaces. *International Journal of Environmental Research and Public Health*, 18(7), 3647. <https://doi.org/10.3390/ijerph18073647>
- [3] Massachusetts General Hospital Center for Women's Mental Health. (November 2021). Essential reads: Who is at risk for depression during the perimenopause? - MGH center for women's mental health. MGH Center for Women's Mental Health - Perinatal & Reproductive Psychiatry at Mass General Hospital. Retrieved May 7, 2024 from <https://womensmentalhealth.org/posts/essential-reads-who-is-at-risk-for-depression-during-the-perimenopause/>

- [4] Sermin Timur and Nevin Hotun Sahin, 2010. The prevalence of depression symptoms and influencing factors among perimenopausal and postmenopausal women. *Menopause*. 17(3), 545-51. doi: <https://doi.org/10.1097/gme.0b013e3181cf8997>.
- [5] Jennifer Gordon and Girdler Susan, 2014. Hormone replacement therapy in the treatment of perimenopausal depression. *Curr Psychiatry Rep*. 16(12):517. <https://doi.org/10.1007/s11920-014-0517-1>.
- [6] Jayashri Kulkarni, 2018. Perimenopausal depression - an under-recognised entity. *Aust Prescr*. 41(6), 183-185. DOI: <https://doi.org/10.18773/austprescr.2018.060>.
- [7] Oura Ring. What makes the Oura Ring Different? Retrieved May 26, 2024 from <https://ouraring.com/blog/the-oura-difference/>
- [8] Oura Ring. How to Use Oura Temperature Trends to Track Your Cycle. Retrieved May 26, 2024 from https://ouraring.com/blog/temperature-to-track-your-menstrual-cycle/?g_acctid=404-975-2641&g_adgroupid=&g_adid=&g_adtype=none&g_campaign=pmax_prospecti ng_integrated_USA_purchase_english_mh&g_campaignid=20915188257&g_k eyword=&g_keywordid=&g_network=x&utm_campaign=OUR_SEM_Spring_Q3FY24_US_Google-PMAX_Search_Various_XDV_Digital_CPC_Various_Various_04-01-2024_06-30-2024_Placement&utm_content=Spring_Q3FY24&utm_id=&utm_medium=cpc &utm_source=google&utm_term=&gad_source=1&gclid=EAlaIqObChMlyPye nJLkhgMVfK5HAR2Psw03EAAYASAAEgJTPvD_BwE&gclsrc=aw.ds
- [9] Janessa Griffith, Olzhas Yessenbayev and Tanzeem Choudhury, (submitted) Digital Interventions for Depression in Peri-menopause: A Scoping Review Using the Behavior Change Techniques Taxonomy. *Journal of Medical Internet Research: Mental Health*.
- [10] John Creswell and Vicki Plano Clark. 2017. *Designing and Conducting Mixed Methods Research* (3rd ed.). SAGE Publications, Thousand Oaks, CA.
- [11] Virginia Braun and Victoria Clarke, 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology*. 3(2), 77–101. DOI: <https://doi.org/10.1191/1478088706qp063oa>
- [12] Mahsa Sheikh, M Qassem, Panicos Kyriacou, 2021. Wearable, Environmental, and Smartphone-Based Passive Sensing for Mental Health Monitoring. *Front Digit Health*. 3,662811. DOI: <https://doi.org/10.3389/fdgh.2021.662811>.
- [13] Mohan Babu, Ziv Lautman, Xiangping Lin, Milan Sobota, Michael Snyder, 2204. Wearable Devices: Implications for Precision Medicine and the Future of Health Care. *Annu Rev Med*. 29(75), 401-415. DOI: <https://doi.org/10.1146/annurev-med-052422-020437>.